

ADVANCE

SIMULATED ADVANCE — targeted greenfield electrolytic tinning line

The simulated Ministry-grade evidence shows no domestic tinplate capability, a sustained target-specification demand above minimum efficient scale, and a greenfield line whose supported economics, national value and competition controls pass after every lower route fails.

SAU-H6-721012 · Targeted greenfield entry (route 7) · Mode: Ministry simulation

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Tin-plated or tin-coated flat-rolled iron or non-alloy steel, width 600 mm or more, thickness less than 0.5 mm

HS H6 / 721012

Unchanged public conclusion
INVESTIGATE — binding constraint unresolved

Target-specification demand 95 kt	Effective qualified capacity 0 kt
Published capability distance D* 0.9 ratio	Minimum effective support S* 137 SAR m

Decision conditions

Secure committed can-maker offtake covering at least the minimum efficient scale before award. [1 more; see the full record](#)

Kill conditions

An incumbent flat-rolled producer commits a tinning line before award. [1 more; see the full record](#)

Evidence passport ledger: [P-COMTRADE-721012-M](#) [P-COMTRADE-721012-X](#)

Snapshot: PUBLIC-SAU-H6-721012-2026-09-12 · As of: 2026-09-12

Supporting evidence pack

1. [Identity and specification boundary](#)
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Identity and specification boundary

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Unchanged public conclusion

INVESTIGATE — binding constraint unresolved

Route not yet determined

No route hypothesis can be preferred from public evidence; resolve the named route evidence before any support decision.

Decision conditions

- No greenfield or financial support recommendation before effective capacity and target specification are resolved.

Kill conditions

- Demand is temporary, re-exported or below efficient scale
- Incumbent expansion already closes the specification-adjusted gap

Next evidence actions

- Saudi tariff-line and invoice description
- Importer/offtaker application and qualification requirements
- Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages

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- Route feasibility, additionality, policy and downside economics evidence for the candidate routes
- Importer/offtaker application and qualification requirements

Product identity

Source field	Value	Unit
HS revision	H6	
HS code	721012	
National tariff line	Unavailable Not established by the supplied evidence.	
Commercial name, English source	Tin-plated or tin-coated flat-rolled iron or non-alloy steel, width 600 mm or more, thickness less than 0.5 mm	
Commercial name, Arabic	منتجات مسطحة من الحديد أو الصلب غير المخلوط، مطلية بالقصدير، بعرض 600 مم أو أكثر وبسماكة أقل من 0.5 مم	
Application boundary	HS 7210.12 identifies tin-plated or tin-coated flat-rolled iron or non-alloy steel of width 600 mm or more and thickness below 0.5 mm; target coating, standard, application and qualification remain unresolved.	
Target specification	Electrolytic tinplate (ETP) can stock for two- and three-piece food and beverage cans SYN-MINISTRY-TINPLATE-001::target specification	
Standard	SASO-ASTM A623/A623M (synthetic design basis) SYN-MINISTRY-TINPLATE-001::target specification	
Alloy	Unavailable Not established by the supplied evidence. SYN-MINISTRY-TINPLATE-001::target specification	
Temper	Unavailable Not established by the supplied evidence. SYN-MINISTRY-TINPLATE-001::target specification	
Application	metal packaging can stock SYN-MINISTRY-TINPLATE-001::target specification	
Zinc coating mass	Unavailable Not established by the supplied evidence. SYN-MINISTRY-TINPLATE-001::target specification	g/m ²

Source field	Value	Unit
Tin coating mass	2.8 5.6 SYN-MINISTRY-TINPLATE-001::target_specification	g/m ²
Thickness	0.17 0.32 SYN-MINISTRY-TINPLATE-001::target_specification	mm
Thickness	Unavailable Not established by the supplied evidence. SYN-MINISTRY-TINPLATE-001::target_specification	µm
Width	700 1000 SYN-MINISTRY-TINPLATE-001::target_specification	mm
Customer qualification required	Yes SYN-MINISTRY-TINPLATE-001::target_specification	

Classification revision and concordance

Unavailable

Specification source spans

Unavailable

Demand and public trade signals

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Demand conclusion

Gross imports are a screening signal, not demand at the target specification.

Source field	Value	Unit
Base demand	104 SYN-MINISTRY-TINPLATE-001::demand	kt
Target-specification demand	95 SYN-MINISTRY-TINPLATE-001::demand	kt

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Source field	Value	Unit
Downside demand	80 SYN-MINISTRY-TINPLATE-001::demand	kt
Committed demand	48 SYN-MINISTRY-TINPLATE-001::demand	kt
Announced demand	20 SYN-MINISTRY-TINPLATE-001::demand	kt
Commitment probability	0.7 SYN-MINISTRY-TINPLATE-001::demand	ratio
Observation year	2024	year
Gross import value	115.370525	USD m
Gross import quantity	104.575028	kt
Named buyers	Unavailable Not established by the supplied evidence.	
Buyer concentration	Unavailable Not established by the supplied evidence.	
Demand timing	Unavailable Not established by the supplied evidence.	
Year	Import value, USD m	Import quantity, kt
2022	188.866936	Unavailable
2023	87.175117	61.390945
2024	115.370525	104.575028

Trade coverage and measurement

Domestic-origin exports separated	No
Execution ceiling	DEGRADED where continuity or tariff-line detail is required; 2021 (H5) excluded by the latest-revision H6 rule Partner detail: PARTNER_DETAIL_OBSERVED (P-WITS-721012-PARTNERS).
Trade-flow basis	gross
Missing years	None recorded; no completed review is implied.
Monthly partner/tariff-line detail available	No

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Quantity is comparable	No
Re-exports separated	No

Production, retained flows and re-exports

Domestic-origin exports (kt)	Unavailable
Domestic production (kt)	Unavailable
Reference year	2024
Re-exports (kt)	Unavailable
Retained imports (kt)	Unavailable
Source evidence references	None recorded; no completed review is implied.

Supplier diagnostics

Partner detail (2024 imports)

Observed from wits_trade — 12 partner rows

Acquisition attempt references	None recorded; no completed review is implied.
Observed partner rows	12
Observed evidence passport	P-WITS-721012-PARTNERS
Partner snapshot identifier	PARTNERS-SAU-WITS-TRADE-2026-09-12
Source explanation	Unavailable
Source identifier	wits_trade
State	PARTNER_DETAIL_OBSERVED
Recorded unit	721012 - imports 2024

Domestic supply and qualified capacity

Supply conclusion

Public producer observations; nameplate is not qualified capacity.

Named producer	Value
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No named producer evidence is available.

Observed adjacency signals

- None recorded; no completed review is implied.

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Source field	Value	Unit
Simulated nameplate capacity	0 SYN-MINISTRY-TINPLATE-001::plant_line	kt
Availability factor	0 SYN-MINISTRY-TINPLATE-001::plant_line	ratio
Yield factor	0 SYN-MINISTRY-TINPLATE-001::plant_line	ratio
Qualification share	0 SYN-MINISTRY-TINPLATE-001::plant_line	ratio
Market allocation share	0 SYN-MINISTRY-TINPLATE-001::plant_line	ratio
Current utilisation	Unavailable Not established by the supplied evidence. SYN-MINISTRY-TINPLATE-001::plant_line	ratio
Description	Targeted greenfield electrolytic tinning line (cold-rolled substrate sourced domestically or imported) SYN-MINISTRY-TINPLATE-001::upgrade	
Incremental capacity	90 SYN-MINISTRY-TINPLATE-001::upgrade	kt
Upgrade schedule	36 SYN-MINISTRY-TINPLATE-001::upgrade	months
Effective qualified capacity	0	kt
Qualified available supply	0	kt

Gap diagnosis and false-positive controls

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Timing gap

Source field	Value	Unit
Specification-adjusted gap	95	kt

Source reference	Status	Value
EX-01_HETEROGENEOUS_RESIDUAL	Not satisfied	The commercial product cannot be separated from a highly heterogeneous residual code and no product-level evidence is available. SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs
EX-02_MARKET_BELOW_MES	Not satisfied	The sustainable downside market is below minimum efficient scale and no credible export contract exists. SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs
EX-03_UNSATISFIABLE_HARD_GATE	Not satisfied	A legal, safety, environmental, IP or customer-qualification hard gate cannot be satisfied. SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs
EX-04_IDLE_EQUIVALENT_CAPACITY	Not satisfied	Domestic production meets the required specification with material idle capacity and no binding market failure is evidenced. SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs

Source reference	Status	Value
EX-05_TRANSITORY_OR_MEASUREMENT	Not satisfied	The apparent gap is dominated by re-export, one-off project demand, temporary price arbitrage or classification discontinuity. SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs
EX-06_REDUNDANCY_OR_CROWD_OUT	Not satisfied	The intervention would create unacceptable redundant capacity or crowd out a more efficient incumbent. SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs

Test the apparent gap

Source reference	Status	Value
hard_exclusion	Not satisfied	HARD_EXCLUSION_NOT_SATISFIED SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs
false_or_measurement_gap	Not satisfied	FALSE_OR_MEASUREMENT_NOT_ESTABLISHED SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs
equivalent_idle_qualified_supply	Not satisfied	EQUIVALENT_QUALIFIED_SUPPLY_NOT_ESTABLISHED SYN-MINISTRY-TINPLATE-001::demand SYN-MINISTRY-TINPLATE-001::equivalence
uneconomic_at_efficient_scale	Not satisfied	ECONOMIC_ROUTE_EXISTS SYN-MINISTRY-TINPLATE-001::route_evidence
structural_overcapacity	Not satisfied	STRUCTURAL_OVERCAPACITY_NOT_ESTABLISHED SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs

Source reference	Status	Value
generic_capacity_contradicted	Not satisfied	GENERIC_CAPACITY_NOT_CONTRADICTED P-COMTRADE-721012-X

Capability and route alternatives

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Source field	Value	Unit
Known coverage K	1	ratio
Unknown weight U	0	ratio
Uncertainty penalty (lambda)	0.5	ratio
Minimum known coverage	0.7	ratio
Published capability distance D*	0.9	ratio

Source field	State	Weight
Feedstock chemistry	2	0.05
Core process route	3	0.2
Equipment operating envelope	3	0.2
Finishing and specification control	3	0.15
Quality, laboratories and metrology	2	0.1
Certification and customer qualification	3	0.1
Capacity and timing	3	0.1

Source field	State	Weight
Utilities, environment and permitting	1	0.05
Skills and market integration	2	0.05

Sector-profile hard gates

Substrate range	RESOLVED
Width/thickness envelope	RESOLVED
Coating process and mass	RESOLVED
Surface treatment	RESOLVED
Mandatory or customer standard	RESOLVED

Unresolved hard gates • None recorded; no completed review is implied.

Known hard-gate failures • None recorded; no completed review is implied.

No intervention

Fails

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Fails
Additionality	Passes
Policy permissibility	Passes

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

passes

- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

Administrative or regulatory correction

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

Information and market linkage

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
-

Certification, testing and quality

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

Demand aggregation and offtake

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

Brownfield upgrade or expansion

Fails

Source field	Value
Feasibility	Fails
Resolves the binding constraint	Passes
Additionality	Passes
Policy permissibility	Passes

Unsupported NPV -136.839

Unsupported IRR 0.06986

Minimum effective support S* 137

Lower-route precedence: No lower route recorded

Incremental national value	220
Capacity ratio	1.125
Warning threshold	1.25
Passes the recorded test	Yes

- The route is not technically feasible on the available evidence.

[SYN-MINISTRY-TINPLATE-001::route_evidence](#)

Technology partnership or joint venture

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value	Not calculable
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Not calculable

- The detected binding constraint is outside this route's intervention class.

Greenfield capacity

Passes

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Passes
Additionality	Passes
Policy permissibility	Passes
Unsupported NPV	-136.839
Unsupported IRR	0.06986
Minimum effective support S*	137

Lower-route precedence: No lower route recorded

Incremental national value	220
Capacity ratio	1.125
Warning threshold	1.25
Passes the recorded test	Yes

[SYN-MINISTRY-TINPLATE-001::route_evidence](#)

Shared enabling infrastructure

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- GRAPH_REQUIRED: route 8 requires the governed dependency graph and a positive UnlockValue.

Public route comparison

No intervention

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

Administrative or regulatory correction

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- Administrative, classification or regulatory barrier evidence is required.

Information and market linkage

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified information, linkage or matching constraint is required.

Certification, testing and quality

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified specification, certification, testing or qualification constraint is required.

Demand aggregation and offtake

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable
Lower-route precedence: No lower route recorded	
Incremental national value	Not calculable
Not calculable	
A verified demand-fragmentation or offtake constraint is required.	

Brownfield upgrade or expansion

Not calculable	
Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable
Lower-route precedence: No lower route recorded	
Incremental national value	Not calculable
Not calculable	
Verified incumbent adjacency makes brownfield the priority hypothesis; effective capacity and economics remain required.	

Technology partnership or joint venture

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified technology constraint and major-line/JV adjacency are required.

Greenfield capacity

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No lower route recorded

Incremental national value	Not calculable
Not calculable	
Greenfield requires failed lower routes, D* above the configured band, demand at MES, and passing competition gates.	

Shared enabling infrastructure

Not calculable	
Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value	Not calculable
Not calculable	
GRAPH_REQUIRED: route 8 requires the governed dependency graph and a positive UnlockValue.	

Counterfactual

Can the incumbent meet the specification without capital investment?	No
Does downside demand support both incumbent and entrant?	Unavailable
Does entry displace efficient domestic production?	No

Is a technology partnership superior to expansion or greenfield?	No
Missing capabilities	• feedstock_chemistry• core_process_route • equipment_envelope• finishing_spec_control • qa_lab_metrology• certification_customer_qualification • capacity_time_window• utilities_ehs_permitting • skills_market_integration
Brownfield versus greenfield comparison	
Incremental capacity	90
Upgrade schedule	36
Unsupported NPV	-136.839
Minimum effective support S*	137
Greenfield alternative	Targeted greenfield electrolytic tinning line at minimum efficient scale.

Economics and intervention

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Source field	Value	Unit
Unsupported NPV	-136.839	SAR m
Unsupported IRR	0.06986	ratio
Minimum effective support S*	137	SAR m
Supported NPV	0.161	SAR m
Supported IRR	0.12007	ratio
Hurdle rate	0.12	ratio

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
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Source field	Value	Unit
Separate downside result	Not calculable Cannot be calculated from the supplied evidence.	SAR m

National-value comparison

National benefits	350
National costs	130
Incremental national value	220
Positive result	Yes

Components

Displacement	15
Domestic value added	210
Exports	30
Fiscal receipts	40
Government cost	60
Knowledge and skills	25
Resilience value	45
Resources and environment	20
Risk allowance	35

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

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Cash flows without support (SAR m), in source order	-620 • 78 • 78 • 78 • 78 • 78 • 78 • 78 • 78 • 78 • 78 • 78 78
Currency	SAR_m
Hurdle rate	0.12
Minimum efficient scale (kt)	60

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
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National value

Displacement	15
Domestic value added	210
Exports	30
Fiscal receipts	40
Government cost	60
Knowledge and skills	25
Resilience value	45
Resources and environment	20
Risk allowance	35

Support instrument	conditional capex co-investment with sunset and clawback
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[SYN-MINISTRY-TINPLATE-001::economics](#)

Cash-flow inputs in source order

• Route code	5
Recorded basis	No incumbent electrolytic tinning line exists to debottleneck or expand.
• Route code	7
Downside cash flows (SAR m), in source order	◦ -620◦ 78◦ 78◦ 78◦ 78◦ 78◦ 78◦ 78◦ 78◦ 78◦ 78◦ 78 ◦ 78
Hurdle rate	0.12
Recorded basis	Targeted greenfield electrolytic tinning line at minimum efficient scale.

[SYN-MINISTRY-TINPLATE-001::route evidence](#)

Competition and public policy

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Source field	Value	Unit
Capacity to downside demand ratio	1.125	ratio
Displacement cost	15	SAR m
Sunset date	Unavailable Not established by the supplied evidence.	
Capacity to downside demand ratio	1.125	
Warning threshold	1.25	
Warning is triggered	No	
Triggers the default warning	Yes	
Displacement cost	15	
Non-additionality assessment		
Status	passes	
Recorded basis	route_record_additionality	
Concentration before and after	Not calculable	

Calculation and recommendation do not authorize public support.

Complete rule execution ledger

R0 · Identity gate

Full · Fires

HS revision H6 and HS6 721012 are frozen in the snapshot; specification/application remains separately gated.

Continue to screening while preserving the unresolved decision object.

R1-F · Persistent retained exposure — full

Full · Does not fire

Full persistence test did not fire.

Do not claim the full persistence rule.

Missing years	• None recorded; no completed review is implied.
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R1-D · Persistence signal — degraded

Degraded · Fires

3 positive observed years within the configured window; no interpolation used.

Generate INVESTIGATE only; confidence capped at C.

Positive observed years	• 2022• 2023• 2024
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Confidence ceiling	C
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R2 · Quantity-led expansion

Full · Fires

Quantity-led expansion signal fires.

Separate structural volume growth from price movement.

First year	2023
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Final year	2024
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Observed period (years)	1
Change in log value	0.2802
Change in log quantity	0.5326
Change in log unit value	-0.2524
Quantity contribution share	0.6785
Quantity compound annual growth rate	0.7034

R3 · Supplier concentration

Disabled · Not evaluable

Value- and quantity-basis partner concentration are NOT_CALCULABLE.

No concentration inference.

Concentration index (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier share	Not calculable
Concentration threshold (HHI)	0.25
Largest-supplier threshold	0.5
Value	

Recorded basis	value
Status	Not calculable
Observation year	2024
Trade-flow basis	gross
Source	partner_observations
Concentration index (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier share	Not calculable
Valid observation coverage	1
Eligible partner count	12

Source explanation	Eligible partner rows do not reconcile to the matching world aggregate value basis.
Quantity	
Recorded basis	quantity
Status	Not calculable
Observation year	2024
Trade-flow basis	gross
Source	partner_observations
Concentration index (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier share	Not calculable
Valid observation coverage	1
Eligible partner count	12
Source explanation	Eligible partner rows do not reconcile to the matching world aggregate quantity basis.

R4-F · Product-tier heterogeneity — full

Disabled · Not evaluable

Partner-month × tariff-line cells are unavailable.

Do not fit clusters or claim grades.

R4-D · Unit-value dispersion — degraded

Degraded · Fires

Comparable annual partner unit values show descriptive dispersion; no cluster or grade conclusion.

Open specification research only; no grade conclusion.

Status	CALCULABLE
Calculation basis	partner_observations
Valid value coverage	1

Valid quantity coverage	1
Minimum valid-value coverage	0.7
Weighted first quartile (USD/t)	1065.9099
Weighted median (USD/t)	1109.4056
Weighted third quartile (USD/t)	1109.4056
Interquartile range (USD/t)	43.4957
Bulk unit-value band (USD/t)	Not calculable
Potential outlier	
Trading partner	United States
Unit value (USD/t)	102222.2222
Quantity share	0
Confirmed outlier	No
Source explanation	Unavailable

R5 · Domestic supply plus continued imports

Disabled · Not evaluable

Domestic capability or a usable import measure is unavailable.

Test specification, qualification, capacity, price, application and allocation mismatch.

Retained imports (kt)	Not calculable
Net import exposure (kt)	Not calculable
Apparent consumption (kt)	Not calculable
Retained imports / apparent consumption	Not calculable
Calculation basis	Not calculable
Unavailable inputs	- domestic_origin_exports_kt• domestic_production_kt - reexports_kt• retained_imports_kt
Reasons for unavailability	
Retained imports (kt)	reexports_kt• retained_imports_kt

Net import exposure (kt)	- domestic_origin_exports_kt - retained_imports_kt
Apparent consumption (kt)	- domestic_origin_exports_kt - domestic_production_kt - retained_imports_kt
Retained imports / apparent consumption	- apparent_consumption_kt - retained_imports_kt
Configured threshold	0.2
Source explanation	Domestic production quantity and retained-import flow are absent from the frozen public snapshot; gross imports cannot establish apparent consumption.

R6 · Capacity pressure

Disabled · Not evaluable

Effective utilisation and target-specification shortage are not public.

Obtain line-level availability, yield, qualification share and allocation.

R7 · Latent domestic capacity

Disabled · Not evaluable

Idle qualified capacity and specification equivalence are not established publicly.

Do not assume latent capacity.

R8 · Committed future demand

Disabled · Not evaluable

Awarded/financed demand at the target specification is unavailable.

Keep base, committed and announced demand separate when provided.

R9-S · Coarse incumbent adjacency screen

Disabled · Not evaluable

No defensible coarse adjacency signal.

Open the full line-level capability assessment; do not publish D* from screening alone.

Same process family Not calculable

Qualifying signal count	0
Known hard-gate failure	No

R10 · Strategic criticality

Disabled · Not evaluable

No responsible-authority criticality designation is present in the demo snapshot.

Keep commercial viability and strategic value separate.

Criticality status	Not calculable
Criticality evidence reference	Unavailable

R11 · Economic exclusion / generic-capacity warning

Degraded · Does not fire

Export/import ratio or established nameplate capability is NOT_CALCULABLE; no generic-capacity exclusion fires.

Reject generic support or move only a named specialty exception to investigation.

Export/import value ratio	0
Export/import ratio threshold	50
Calculated export/import value ratio	0
Disclosed export/import value ratio	Unavailable
Disclosed ratio is consistent	Unavailable
Ratio basis	gross_trade_value
Ratio status	CALCULABLE
Ratio explanation	Unavailable
Established nominal capacity	
Established by evidence	No
Total observed nominal capacity (t/year)	Not calculable

Named producers	• None recorded; no completed review is implied.
Evidence references	• None recorded; no completed review is implied.
Source explanation	Verified domestic capability is not established.

R12 · Evidence-value trigger

Degraded · Fires

5 named facts could change the route.

Prioritise the smallest evidence request with a plausible route effect; quantify EVSI when cost and value inputs exist.

Named missing facts	• Saudi tariff-line and invoice description • Importer/offtaker application and qualification requirements • Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages • Route feasibility, additionality, policy and downside economics evidence for the candidate routes • Importer/offtaker application and qualification requirements
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Evidence needed

- Saudi tariff-line and invoice description

Evidence-need identifier	identity/tariff-line
Blocked decision field	product_identity
Effect on route	Can resolve the decision object and change the admissible route.
Evidence references	◦ P-WCO-721012
Numerical value of information	Not calculable

- Importer/offtaker application and qualification requirements

Evidence-need identifier	target specification/application
Blocked decision field	demand_at_required_specification
Effect on route	Can establish target demand and change state or route.
Evidence references	◦ None recorded; no completed review is implied.

Numerical value of information	Not calculable
Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages	
Evidence-need identifier	capacity/availability/allocation
Blocked decision field	domestic_supply_or_capability
Effect on route	Can distinguish no action, brownfield expansion, and new capacity.
Evidence references	None recorded; no completed review is implied.
Numerical value of information	Not calculable
Route feasibility, additionality, policy and downside economics evidence for the candidate routes	
Evidence-need identifier	route economics
Blocked decision field	route_economics
Effect on route	Can determine whether the plausible route proceeds unsupported.
Evidence references	None recorded; no completed review is implied.
Numerical value of information	Not calculable
Importer/offtaker application and qualification requirements	
Evidence-need identifier	line-level production or producer-grade matrix
Blocked decision field	hard_regulatory_or_process_gate
Effect on route	Can establish domestic equivalence and change the incumbent route.
Evidence references	None recorded; no completed review is implied.
Numerical value of information	Not calculable

Simulated R6–R8 ledger

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

R5 · Retained import penetration

Disabled · Not evaluable

Synthetic retained-flow inputs are absent; R5 is not calculable.

Use only when production and retained-flow blocks are declared.

Retained imports (kt)	Not calculable
Domestic production (kt)	Not calculable
Gross import quantity	104.575028
Retained-import penetration	Not calculable
Apparent consumption (kt)	Not calculable
Minimum retained-import penetration	0.2

R6 · Capacity pressure

Disabled · Not evaluable

Synthetic utilisation or effective qualified capacity is unavailable; R6 is not calculable.

Use as a DEGRADED capacity-pressure signal only; it cannot by itself support ADVANCE without a sustained period.

Effective utilisation	Not calculable
Minimum effective utilisation	0.85
Target-specification demand	95
Effective qualified capacity	0
Effective-capacity source field	effective_qualified_capacity_kt
Shortage ratio	Not calculable
Shortage denominator	effective_qualified_capacity_kt

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Minimum specification-matched shortage	0.1
Sustained period	Not calculable

R7 · Latent domestic capacity

Disabled · Not evaluable

Synthetic latent capacity is not calculable because a required utilisation or equivalence input is absent.

Test no-support, market linkage, procurement, or barrier removal only when equivalence and latent capacity are established.

Effective utilisation	Not calculable
Maximum effective utilisation	0.7
Specification equivalence	No
Qualified available supply	Not calculable

R8 · Committed future demand

Full · Fires

Committed and announced layers are disclosed, but base demand, commitment probability, and minimum efficient scale govern R8 calculability.

Do not infer probability-adjusted demand addition or MES fill; keep committed and announced demand separate.

Base demand	104
Committed demand	48
Announced demand	20
Target-specification demand	95
Downside demand	80
Commitment probability	0.7
Probability-adjusted committed demand (kt)	33.6
Probability-adjusted demand addition	0.323077

Minimum probability-adjusted demand addition	0.2
Minimum efficient scale (kt)	60
Minimum efficient scale coverage	0.56
Minimum scale-coverage threshold	0.25

Evidence passports and contradictions

4 public records; 14 synthetic records.

Contradiction register

Public evidence contradictions

No public contradictions recorded.

Synthetic evidence contradictions

No synthetic contradictions are recorded for this simulation.

P-COMTRADE-721012-M

UN Comtrade Saudi imports rows for HS 721012

Source field	Value
Source	UN Comtrade
Class	B
Status	calculated
Source period	2022/2023/2024
Retrieval date	2026-09-12
Reviewer status	unconfirmed_by_responsible_authority
Transformation	LATEST_REVISION_ONLY H6 rows; trade value divided by 1,000,000 to USD million and net weight divided by 1,000,000 to kt.
Supporting claims	• TRADE_QUANTITY• TRADE_VALUE

Source field	Value
Contradiction register	Unavailable
Source reference	https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&period=2024&flowCode=M&cmdCode=AG6&partnerCode=0&motCode=0&customsCode=C00

P-COMTRADE-721012-X

UN Comtrade Saudi exports rows for HS 721012

Source field	Value
Source	UN Comtrade
Class	B
Status	calculated
Source period	2022/2023/2024
Retrieval date	2026-09-12
Reviewer status	unconfirmed_by_responsible_authority
Transformation	LATEST_REVISION_ONLY H6 rows; trade value divided by 1,000,000 to USD million and net weight divided by 1,000,000 to kt.
Supporting claims	• EXPORT_IMPORT_RATIO• TRADE_QUANTITY• TRADE_VALUE
Contradiction register	Unavailable
Source reference	https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&period=2024&flowCode=X&cmdCode=AG6&partnerCode=0&motCode=0&customsCode=C00

P-WCO-721012

HS Nomenclature 2022 — Chapter 72

Source field	Value
Source	wco_hs_nomenclature
Class	B
Status	observed
Source period	2022
Retrieval date	2026-09-12

Source field	Value
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Verbatim span from DocumentRecord DOC-WCO-HS-NOMENCLATURE-5530bdf97060-8da42b5156a2.
Supporting claims	TARGET_PRODUCT_IDENTITY
Contradiction register	Unavailable
Source reference	https://www.wcoomd.org/-/media/wco/public/global/pdf/topics/nomenclature/instruments-and-tools/hs-nomenclature-2022/2022/1572_2022e.pdf?la=en

P-WITS-721012-PARTNERS

WITS Saudi 2024 import partner rows for HS 721012

Source field	Value
Source	World Bank WITS
Class	B
Status	calculated
Source period	2024
Retrieval date	2026-09-12
Reviewer status	unconfirmed_by_responsible_authority
Transformation	World aggregate excluded; value divided by 1,000,000 to USD million and net weight divided by 1,000,000 to kt. Partner detail state: PARTNER_DETAIL_OBSERVED (wits_trade; 12 partner rows).
Supporting claims	SUPPLIER_CONCENTRATION
Contradiction register	Unavailable
Source reference	https://wits.worldbank.org/trade/comtrade/en/country/SAU/year/2024/tradeflow/Imports/partner/ALL/product/721012

SYN-MINISTRY-TINPLATE-001::capability_states

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Capability States

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: capability_states
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::class_if_confirmed

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Class If Confirmed

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable

Source field	Value
Supporting claims	Simulation branch input: class_if_confirmed
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::counterfactual

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Counterfactual

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: counterfactual
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::decision_specific_hard_gates

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Decision Specific Hard Gates

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: decision_specific_hard_gates
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::demand

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Demand

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: demand
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::economics

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Economics

Source field	Value
Source	DEMO_GENERATOR

Source field	Value
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: economics
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::equivalence

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Equivalence

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: equivalence
Contradiction register	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Evsi

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: evsi
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::hard_exclusion_inputs

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Exclusion Inputs

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable

Source field	Value
Supporting claims	Simulation branch input: hard_exclusion_inputs
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::hard_gates

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Gates

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: hard_gates
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::plant_line

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Plant Line

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: plant_line
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::route_evidence

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Route Evidence

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: route_evidence
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::target_specification

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Target Specification

Source field	Value
Source	DEMO_GENERATOR

Source field	Value
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: target_specification
Contradiction register	Unavailable

SYN-MINISTRY-TINPLATE-001::upgrade

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Upgrade

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: upgrade
Contradiction register	Unavailable

Conditions, kill conditions and next facts

Class D simulation · SYN-MINISTRY-TINPLATE-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Decision conditions

- Secure committed can-maker offtake covering at least the minimum efficient scale before award.
- Support is capped at the calculated minimum, milestone-based and subject to clawback.

Kill conditions

- An incumbent flat-rolled producer commits a tinning line before award.
- Downside demand falls below minimum efficient scale.

Next evidence actions

- None

Source field	Value	Unit
Approximate EVSI	111	SAR m
Route-change probability	0.55	
Value difference (SAR m)	220	
Evidence acquisition cost (SAR m)	4	
Delay cost (SAR m)	6	
Approximate EVSI	111	
Positive result	Yes	
Next fact to establish	Confirm committed can-maker offtake and substrate sourcing	

Decision history and expert overrides

None recorded; no completed review is implied.

Authority and reproducibility

Authority and versions

Source reference	Value
Methodology	docs/authority/Industrial_Opportunity_Resolution_Methodology_Final_KSA.docx 5717cbd42acc9947ce5e450013719275acb7ed1470847b21fb2cc547c8ac4ce9
Project	0.2.0
Thresholds	1.3.0
Sector profiles	1.1.0
Evidence policy	1.4.0
UI strings	1.7.0

Evidence boundary

Public decision uses public evidence only	Yes
Synthetic branch isolation	Yes
Scenario reconciliation	
Status	PASS
Scenario identifier	SYN-MINISTRY-TINPLATE-001
Opportunity identifier	SAU-H6-721012
Public snapshot identifier	PUBLIC-SAU-H6-721012-2026-09-12
Recorded checks	

• Rule identifier	target_spec_demand_within_public_imports
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	target_spec_demand_kt <= latest_public_imports_kt
Recorded result	PASS
Blocking check	Yes

Source inputs	
Target-specification demand	95
Latest public observation year	2024
Latest public imports (kt)	104.575028
Source explanation	95 kt <= 104.575 kt.
• Rule identifier	line_nameplate_within_disclosed_public_capacity
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	plant_line.nameplate_kt <= sum(non_null installed_capacity_tpy) / 1000
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Line nominal capacity (kt)	0
Disclosed installed capacity (t/year)	◦ None recorded; no completed review is implied.
Publicly disclosed nominal capacity (kt)	Unavailable
Expansion assumption applied	No
Source explanation	No numeric public installed nameplate is disclosed.
• Rule identifier	capacity_factors_within_unit_interval
Source	Core 06 §10 item 3 lines 226-233; Core 06 §6.2 lines 148-156; Core 05 §8 lines 197-203
Recorded formula	for every capacity factor: 0 <= factor <= 1
Recorded result	PASS
Blocking check	Yes
Source inputs	
Availability factor	0
Yield factor	0

Qualification share	0
Market allocation share	0
Source explanation	All declared capacity factors are within [0,1].
• Rule identifier	qualified_availability_within_physical_output
Source	Core 06 §5.1 lines 92-101; Core 06 §10 item 3 lines 226-233
Recorded formula	qualified_available_kt <= nameplate_kt * availability * yield
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No qualified_available_kt is declared.
• Rule identifier	demand_layers_remain_separate
Source	Core 04 §2.6 lines 102-112
Recorded formula	observe downside_demand_kt <= target_spec_demand_kt and committed_demand_kt <= target_spec_demand_kt; no decision effect
Recorded result	INFORMATIONAL
Blocking check	No
Source inputs	
Target-specification demand	95
Downside demand	80
Committed demand	48
Base demand	104
Downside demand within target demand	Yes
Committed demand within target demand	Yes
Source explanation	Demand layers are reported separately; no ordering constraint is enforced.

• Rule identifier	tariff_line_allocation_sums_to_public_hs6_total
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	sum(tariff_line_allocation.lines.quantity_kt) == latest public imports_kt
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No tariff_line_allocation block is declared.
• Rule identifier	buyer_allocation_within_public_imports
Source	Core 06 §5.1 lines 92-101
Recorded formula	sum(buyer_allocation.buyers.quantity_kt) <= latest public imports_kt
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No buyer_allocation block is declared.
• Rule identifier	expansion_assumption_disclosed_and_bounded
Source	Core 06 §5.1 lines 92-101; KL-38
Recorded formula	planned_nameplate_kt >= disclosed public nameplate; commissioning_year >= public as_of year; plant_line.nameplate_kt <= planned_nameplate_kt
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No expansion_assumption block is declared.
• Rule identifier	retained_flows_reconcile_to_public_trade
Source	Core 06 §5.1; methodology §3.3

Recorded formula	retained_imports_kt <= imports_kt; retained + reexports == imports when reexports numeric; domestic_origin_exports_kt <= public exports_kt
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No production_and_retained_flows block is declared.
• Rule identifier	base_demand_and_commitment_probability_valid
Source	Core 06 §5.2; Core 07 §3 R8
Recorded formula	0 <= commitment_probability <= 1; base_demand_kt > 0; minimum_efficient_scale_kt > 0 when declared
Recorded result	PASS
Blocking check	Yes
Source inputs	
Base demand	104
Commitment probability	0.7
Minimum efficient scale (kt)	60
Source explanation	Base demand, commitment probability and MES inputs are valid.
• Rule identifier	shared_enabler_declaration_valid
Source	Core 06 scenario contract 2.1.0 and methodology §8.3
Recorded formula	explicit route valuation and complete Class-D declaration
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No shared-enabler declaration is present.

Latest imports (USD million)	115.370525
Latest imports (kt)	104.575028
Public decision unchanged after simulation	Yes
Scenario expected-result comparison	
Expected result	
State	ADVANCE
Route code	7
Actual result	
State	ADVANCE
Route code	7
Results match	Yes